Certificate-checked simple connectivity of a surface-bundle surgery manifold

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Abstract

We define a compact marked surface-bundle surgery manifold V_{aud} and prove that $\pi_1(V_{\mathrm{aud}}) = 1$. Four explicit finite presentations are trivial under a published derivation-certificate specification; a geometric reconstruction identifies one of them with $\pi_1(V_{\mathrm{aud}})$. The same construction gives $\chi(V_{\mathrm{aud}}) = 2$, integral homology concentrated in degrees 0 and 2, spinness, a primitive square-zero genus-two fiber, and a surviving relative section. A separate framing theorem identifies the product push-offs used in the definition with canonical Lagrangian-framing classes, so standard Luttinger surgery gives a symplectic structure on V_{aud}. Automated certificate replay assumes that at least one checker conforms to the mathematical specification; the two shipped implementations have not received an independent human line-by-line audit. The construction was motivated by Wuebben's proposed exotic $S^2 \times S^2$, but identifying V_{aud} with Wuebben's fixed member remains an open comparison problem, recorded clause by clause in the supplement. No exotic-manifold conclusion is asserted here.

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1 Introduction

This paper defines a compact marked surface-bundle surgery manifold V_{aud} and proves

$$\pi_1(V_{\mathrm{aud}}) = 1.$$

The proof has two independent-looking but logically consecutive parts. Finite derivation certificates first prove that four explicit presentations are trivial. A geometric argument then reconstructs the marked bundle, torus exterior, peripheral curves, and drilled transport relation, and identifies one fixed presentation with $\pi_1(V_{\mathrm{aud}})$. The other three presentations are algebraic robustness checks, not additional geometric manifolds. Elementary topology and a separate framing theorem then give the intrinsic homology, spin, fiber, section, and symplectic statements in Theorem 2.2.

The construction was motivated by the disputed fundamental-group step in Wuebben's proposed exotic-manifold construction [1]. Simple connectivity of Wuebben's piece V is an input to the proposed exotic $S^2 \times S^2$, quotient, and $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$ conclusions. A contrary computation attributed to Lidman and Piccirillo has been reported, but no presentation or nontriviality witness for it is publicly available in the materials examined here. This paper does not resolve that dispute: it proves a theorem about the separately defined V_{aud} and records the remaining source-to-audit comparison as an open checklist in the supplement.

Triviality is deliberately not used as evidence that a proposed source-to-presentation dictionary is correct. Wuebben's public computations report trivial groups for all 72 cases in one nearby grid and classify 14 of 15 single-word perturbations as SILENT at the group-decision layer [2]; the separate 100-case meridian-shift scan in this project likewise reports only trivial groups. These experiments show that triviality is non-discriminating in this neighborhood. They do not support the source comparison. Instead, the identification of the fixed presentation with V_{aud} is proved by the geometric reconstruction in Sections 4–5, and the further transfer to Wuebben remains an open comparison problem.

The logical architecture is summarized below. Solid arrows are proved in this article; the dashed arrow is the open comparison with Wuebben.

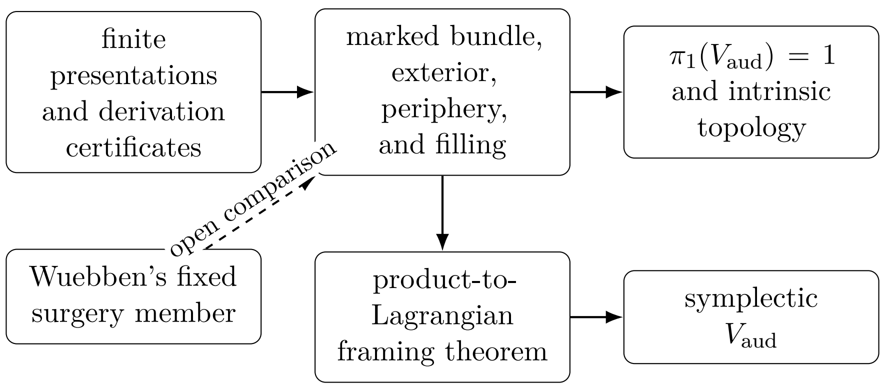


Figure 1: Dependency structure. Certificate checking, geometric conversion, and symplectic framing are different evidence layers.

The frozen transport and theorem-critical certificates replay in separate Python and Ruby implementations. Automated acceptance assumes that at least one checker conforms to the mathematical specification; neither has received an independent human line-by-line audit. The PL and symplectic arguments likewise await external specialist review. These review targets are stated once, in Section 7; implementation inventories, source-comparison clauses, and the conditional downstream history are kept in the companion supplement.

2 The audit manifold and principal statements

Write $\Sigma_{1,1}$ for an oriented once-punctured torus with based oriented spine loops α, β.

Definition 2.1 (the audit manifold)**.** Let F be a closed oriented genus-two surface with an ordered five-chain (a, b, c, d, e) whose regular-neighborhood complement is the disjoint union of two disks containing marked points p, O. Let φ_0 be the orientation-preserving involution reversing the chain, fixing p, O, and acting freely on c by a half-rotation. Let

$$\psi_0 = T_a \circ T_b,$$

with T_b applied first and both twists supported away from a collar of e. Form the marked bundle R_{aud} over $\Sigma_{1,1}$ by clutching the α- and β-mapping cylinders of φ_0 and ψ_0 to one common marked fiber.

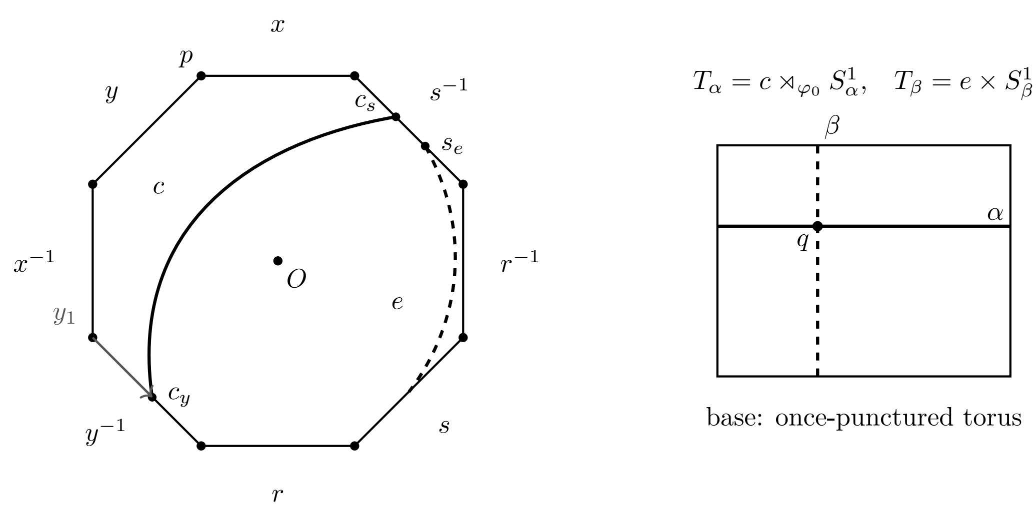
Inside R_{aud} put

$$T_\alpha = c \rtimes_{\varphi_0} S^1_\alpha, \qquad T_\beta = e \times S^1_\beta.$$

Fix the basepoint $q = (p, *)$. Let $A = [\bar{\alpha}]^{-1}$ and $B = [\bar{\beta}]^{-1}$ be the based inverse base loops. Base the oriented meridians M, N along the initial fiber arcs y_1 to c_y and s_2 to s_e, respectively. In the product normal boundaries let λ_α be the y_1-whiskered push-off that lifts the inverse α direction and closes along the half of c running from $c_s = \varphi_0(c_y)$ to c_y, oriented from c_s to c_y, and let λ_β be the s_2-whiskered push-off of the inverse β direction along e. Their based words Ax and $(r^{-1}Mr)B$ are conclusions of Lemmas 5.6 and 5.7, not part of this definition. Define V_{aud} by the product-framed fillings

$$M\lambda_\alpha = 1, \qquad N\lambda_\beta = 1.$$

The orientations of $M, N, \lambda_\alpha, \lambda_\beta$ are part of this definition. Thus this is the $(\epsilon_A, \epsilon_B) = (+1, +1)$ sheet in the fixed audit basis; changing one exponent while holding that basis fixed is not declared to preserve the filling.



(a) the marked fiber F

(b) the two surgery tori

Figure 2: The audit manifold's marked data. Panel (a) is the symmetric octagon model of the genus-two fiber, with boundary word $xyx^{-1}y^{-1}rsr^{-1}s^{-1}$; all eight corners represent p, and O is the center. The marked curves c, e and the initial y_1 whisker determine the curve subbundles and peripheral basing. Panel (b) shows the two tori over the base loops. The drawing is explanatory; the proof uses the certified simplicial paths and incidences.

Theorem 2.2 (Audit-Manifold Theorem)**.** The compact oriented smooth four-manifold V_{aud} has connected boundary, Euler characteristic 2, and

$$\pi_1(V_{\mathrm{aud}}) = 1, \qquad H_k(V_{\mathrm{aud}}; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 0, 2, \\ 0, & k = 1, 3, 4. \end{cases}$$

It is spin. A genus-two fiber F disjoint from the surgery tori survives as a primitive square-zero class generating $H_2(V_{\mathrm{aud}};\mathbb{Z})$, and the fixed point p gives a properly embedded once-punctured-torus section Γ with $F \cdot \Gamma = 1$. Moreover V_{aud} admits a symplectic form for which F is symplectic.

Theorem 2.2 is source-independent: it concerns the explicit object in Definition 2.1. Simple connectivity and the topological assertions are proved at the end of Section 5; symplecticity is proved after the logically separate framing theorem in Section 6.

3 The finite certificate theorem

Theorem 3.1 (finite-certificate statement)**.** Let $\mathcal{M}$ be the manifest identified in the normative-root paragraph below. Its sealed-presentation entry binds a three-generator, seventy-eight-relator presentation Q and four ordered pairs of filling words indexed by $(\epsilon_A, \epsilon_B) \in \{(+,+),(+,-),(-,+),(-,-)\}$. Let $P_{\epsilon_A,\epsilon_B}$ be the presentation obtained from Q by adjoining the corresponding pair. Every one of these four explicitly manifest-bound presentations presents the trivial group.

Normative root. Every file this theorem depends on is bound by the v2.2.2 proof manifest $\mathcal{M}$, whose SHA-256 is

`668e6dac41ac04340c25ca1eb06272512e8ad92353d94c866730dfefd5878d63`

and which lists the digest of each certificate, frozen input, and checker. The manifest path, sealed-input digest, git tree object, archive DOI, and literal replay commands are collected in the supplement. Verifying $\mathcal{M}$ and replaying its named certificates constitutes the complete operational replay of the finite artifact; its mathematical interpretation is fixed by Definitions 3.2–3.4 and Theorem 3.5.

Definition 3.2 (letters, words, and equation keys)**.** Fix generators $g_1, \dots, g_n$. Source relators are signed words in $\{\pm 1, \dots, \pm n\}$, with j denoting g_j. Certificate records use the monoid alphabet $L_n = \{1, \dots, 2n\}$ and the decoding map

$$\mathrm{dec}(2j-1) = j, \qquad \mathrm{dec}(2j) = -j.$$

Thus the required inverse-letter table is $(2, 1, 4, 3, \dots, 2n, 2n-1)$. If w is a signed word, write $\bar{w}$ for its reverse with all signs changed, $\mathrm{fr}(w)$ for free reduction, and $\mathrm{cr}(w)$ for the word obtained by additionally removing inverse first/last pairs until none remain. Define

$$\mathrm{ck}(w) = \min_{\mathrm{lex}}\{\text{cyclic rotations of } \mathrm{cr}(w) \text{ and } \overline{\mathrm{cr}(w)}\},$$

with $\mathrm{ck}(1) = 1$. For certificate words u, v put

$$E(u,v) = \mathrm{ck}\big(\mathrm{dec}(u)\overline{\mathrm{dec}(v)}\big).$$

Equality of equation keys therefore means equality after free and cyclic reduction, cyclic conjugacy, and inversion. Each operation preserves the normal closure of the equation word.

Definition 3.3 (literal rewrite trace)**.** Suppose records $R_0, \dots, R_{i-1}$ have been accepted, with $R_r = (\ell_r, \rho_r)$. A trace on a word w is a list of pairs (r, p) with $r < i$ and $0 \le p \le |w|$. At that step the contiguous subword beginning at p must equal ℓ_r literally, and it is replaced by ρ_r. No cyclic matching, inversion, or implicit free reduction occurs inside a trace. Write $T(w)$ for its final word.

Definition 3.4 (certificate grammar)**.** A `luttinger-kbmag-proof-v1` certificate is a finite ordered list of records (ℓ_i, ρ_i), together with source-binding metadata and identity roots. Every side is a word over L_n. The case index must be an integer in the range of the source filling array. The source digest, case index and slug, generator count, full relator list, and inverse table must agree exactly with the hash-bound input. A record has exactly one of four proof forms.

(i) An *inverse axiom* has $\rho_i = 1$ and $\ell_i = a\,\iota(a)$ for the required inverse table ι.

(ii) An *input-relator record* names an in-range source relator q_k and satisfies $E(\ell_i, \rho_i) = \mathrm{ck}(q_k)$.

(iii) An *overlap record* names parents $a, b < i$, an integer offset d with $-|\ell_b| < d < |\ell_a|$, and two traces. The two parent left sides must agree literally on the resulting nonempty overlap. Let w be the shortest word containing both occurrences. Replacing the occurrence of ℓ_a by ρ_a gives one branch w_a, and replacing ℓ_b by ρ_b gives the other w_b. If the recorded traces give $u = T_a(w_a)$ and $v = T_b(w_b)$, validity requires $E(\ell_i, \rho_i) = E(u, v)$.

(iv) A *change record* names an old record $o < i$ and traces on its two sides. Their literal outputs must equal the separately stored `reduced_left` and `reduced_right` words, say u, v, and validity requires $E(\ell_i, \rho_i) = E(u, v)$.

Finally the root array has length exactly $2n$; its entry for each $a \in L_n$ is an in-range record whose two sides are literally $(a, 1)$. Full-inventory mode also requires one correctly slugged certificate for each of the eight source fillings, with no repeated index or slug and with the stated presentation dimensions.

Theorem 3.5 (certificate soundness)**.** Let G be the group defined by the source relators for a selected filling. If a certificate satisfies Definitions 3.2–3.4, then G is trivial.

Proof. Induct over the records. An inverse axiom is an equality in the free group. For an input record, the equation word is a freely and cyclically reduced cyclic conjugate or inverse of a defining relator, so its two sides agree in G. Every trace replaces a subword by the other side of an earlier valid equation and therefore preserves the represented element of G.

For an overlap, both untraced branches come from one source word by applying one earlier equality, so they agree in G; their traces preserve that equality. Equality of equation keys transfers the resulting equality to (ℓ_i, ρ_i). For a change, the two sides of an earlier valid equation are separately rewritten by earlier equations, and the same equation-key argument applies. Thus every accepted record is an equality in G. The root records say literally that every generator letter and its inverse equal 1, so $G = 1$. □

Definitions 3.2–3.4 are the normative mathematical specification. The complete reference control flow and concrete file format are reproduced in the supplement and in `verification/luttinger/CERTIFICATE_SPEC.md`. The Python and Ruby checkers separately implement it using only their standard libraries. They share neither source code nor runtime, but both were produced within this project and have not received an independent human line-by-line audit.

Proof of Theorem 3.1. For each of the four selected entries, the certificate bundle prepared with this version contains a certificate bound to the manifest-listed source digest, case index, slug, complete relator list, generator count, and inverse table. The Python and Ruby implementations both accept every record and all six identity roots. The corresponding certificate inventory is recorded in the supplement. Hence Theorem 3.5 applies to each selected filling. Separately, two standalone checkers replay the frozen elementary Tietze transport from its serialized raw input to Q. No geometric interpretation of either byte string enters this proof. □

Proposition 3.6 (human-readable terminal collapse)**.** Write the three generators of the fixed $(+,+)$ presentation as a, b, c. Three accepted certificate equations imply the more readable consequences

$$a = bc, \qquad ac^2 = 1, \qquad b^2c = cab.$$

These three consequences already force $a = b = c = 1$.

Proof. Records 6649, 6662, and 6665 of the zero-based ancestry array are, literally,

$$ac^{-1} = b, \qquad c^{-2} = a, \qquad c^{-1}b^2 = abc^{-1}.$$

Right multiplication of the first by c, right multiplication of the second by c^2, and left multiplication of the third by c followed by right multiplication by c give the three displayed consequences. Now $a = bc$ and $ac^2 = 1$ imply $bc^3 = 1$, so $b = c^{-3}$ and $a = c^{-2}$. Substitution into $b^2c = cab$ gives $c^{-5} = c^{-4}$, hence $c = 1$, and then $a = b = 1$. The certificate carries the long proof that the three short equations follow from the original eighty relators; this paragraph exposes the final algebraic reason the group dies. □

The finite theorem is therefore relative only to its published mathematical specification and to the proposition that at least one implementation conforms to it. The rest of the paper asks the different question: what geometric object do the hash-bound presentations describe? The fixed $(+,+)$ presentation is identified with V_{aud} below; comparison with Wuebben is postponed to Section 7. The supplement records the complete implementation inventory and the detailed layer-by-layer trust boundary.

What the three generators mean. The three generators of Q are not privileged loops drawn in the surface-bundle picture. They are the survivors of a proof-producing Tietze transport that starts with one edge generator for each non-tree edge of the drilled simplicial exterior. The seventy-eight relators retain the full two-skeleton information after those eliminations. Geometric meaning is carried separately by eighty-nine tracked words: the original fiber and base loops, both meridians, both product longitudes, and the comparison paths are all expressed in the three survivors. Thus Q is a compact algebraic coordinate system for the torus exterior, while the tracked words are the dictionary back to its geometry. The proof below never assigns a geometric name to a survivor merely because it simplifies a relation.

4 Geometric identification of the audit manifold

This section proves that the $(+,+)$ finite presentation of Theorem 3.1 is a presentation of the explicitly defined V_{aud}. The other three sign sheets remain algebraic robustness checks.

This section makes no comparison with the target paper's denotation; that logically separate problem is discussed in Section 7.

The genus-2 fiber is realized as a regular neighborhood of the specified five-chain of curves — five plumbed annular bands and two cone discs — with the hyperelliptic-type involution φ_0 realized simplicially; its fixed points, the curve actions $a \leftrightarrow e$, $b \leftrightarrow d$, and the free half-rotation of the middle curve c are executable assertions about the complex, not assumptions. The bundle R_{aud} is assembled from simplicial mapping cylinders over the once-punctured torus base; the two surgery tori T_α, T_β arise as induced subcomplexes, and the assembled 4-complex passes manifold checks down to the vertex-link level.

The finite artifact inventory and checker acceptance results are kept in the supplement. They establish finite predicates only. The lemmas below supply the mathematical conversions from simplicial incidences, paths, and word lists to the stated topological conclusions; successful replay is not used as a substitute for those conversions.

The audit dictionary is displayed in Table 1; it is not reconstructed from the fact that the filled groups happen to collapse. Here "literal" means a stored edge path in the triangulation, with its basepoint and orientation retained.

Audit datum	Mathematical role	Finite realization
F, p, O	genus-two fiber with two fixed marks	closed simplicial surface; both fixed vertices pass surface- and link-level tests
a, b, c, d, e	ordered filling five-chain	independently reconstructed ribbon; oriented c at V_2 represents $(rx)^{-1}$
φ_0	$x \leftrightarrow r$, $y \leftrightarrow s$	simplicial half-turn fixing p, O, checked on based paths
$\psi_0 = T_a T_b$	$x \mapsto y^{-1}$, $y \mapsto yx$, $r \mapsto r$, $s \mapsto s$	ordered flip stack with T_b first; based monodromy replay
T_α	$c \rtimes_{\varphi_0} S^1_\alpha$	induced torus on the c-stack; local-flatness and subbundle checks
T_β	$e \times S^1_\beta$	induced product torus on the e-stack; local-flatness and product checks
A, B	$[\bar{\alpha}]^{-1}, [\bar{\beta}]^{-1}$	literal based edge paths from q, retaining their orientations
M, N	meridians based along y_1, s_2	oriented dual normal circles with the same literal whiskers; every normal fiber checked

Table 1: Marked-bundle dictionary for the fixed $n = 0$ member. Peripheral longitudes, which carry the sign-sensitive part of the identification, are displayed separately in Table 2.

Lemma 4.1 (equivariant marked fiber)**.** There is an orientation-preserving homeomorphism $h : |F_K| \to F$ carrying the ordered curves (a, b, c, d, e) and marked points (p, O) to those of Definition 2.1 and satisfying $h\varphi_K = \varphi_0 h$. It carries the named based loops x, y, r, s with their displayed orientations.

Proof. Finite evidence. The independent ribbon reconstruction has

$$i(a, b) = i(b, c) = i(c, d) = i(d, e) = 1, \qquad i(u, v) = 0 \quad \text{otherwise.}$$

Its regular neighborhood retracts to a graph with six vertices and ten edges, hence has Euler characteristic -4. The two disjoint symplectic pairs (a, b) and (d, e) force genus two, so the neighborhood has two boundary components. The complement in the certified genus-two surface has Euler characteristic two and exactly those two boundary circles; it is therefore two disks. An exhaustive rotation-system check begins with the 36 cyclic orders at the independent four-valent crossings. Alternation, orientation preservation, two boundary faces, and preservation of the two faces leave exactly four systems; reversing the permitted curve orientations normalizes all four to one equivariant oriented ribbon type. The primary and independent fibers realize that same labeled type, including the involution on every directed curve end and the two p/O faces.

Topological input. Map vertex disks and edge bands by the checked cyclic correspondence. These product maps agree on attaching intervals, so they give an equivariant homeomorphism of the ribbon neighborhoods. Each complementary disk is invariant: if the involution exchanged them it would have no fixed point there, contrary to the certified two-point fixed set. The periodic-disk classification [6, Theorem 3.1 and Proposition 3.2] identifies each nontrivial orientation-preserving order-two disk action with a half-turn and shows that a periodic disk map fixed on the boundary is the identity. More explicitly, quotient each complementary disk by its half-turn. Both quotients are disks with one marked branch point, and the already constructed equivariant boundary map descends to their boundary circles. The Alexander extension, chosen to carry branch point to branch point, lifts uniquely after one boundary lift is fixed; the lift is equivariant and agrees with the ribbon map on the boundary. This gives the required equivariant marked map relative to the boundary. The based path comparison then reads, literally,

$$\varphi_0 : (x, y, r, s) \longmapsto (r, s, x, y).$$

No fundamental-group filling or downstream theorem enters this argument. □

Lemma 4.2 (relative monodromy and graph clutching)**.** The map h can be chosen so that it conjugates the two monodromy representatives relative to the full c and e collars. It consequently extends to an orientation-preserving fiberwise homeomorphism

$$H : |K| \longrightarrow (R_{\mathrm{aud}})_{\mathrm{top}}$$

carrying the basepoint, both torus subbundles, and their product collars to the corresponding marked objects.

Proof. Finite evidence. The literal based actions are

$$\varphi_0 : (x, y, r, s) \mapsto (r, s, x, y), \qquad \psi_0 : (x, y, r, s) \mapsto (y^{-1}, yx, r, s).$$

The second action is obtained from complete whiskered paths, not free homotopy classes, and is replayed in two differently subdivided beta stacks. On the three-row c collar the alpha map is exactly the free shift by four columns. The beta representative is the ordered supported word $T_a \circ T_b$, with T_b first; every trace cell avoids the full e collar, and the restricted tetrahedra form exactly its product staircase.

Topological conversion. Both bundle models are the union of a common fiber block and two mapping cylinders over the rank-two spine of $\Sigma_{1,1}$; there is no base two-cell. For a monodromy f, write

$$M_f = (F \times [0, 1])/((z, 1) \sim (f(z), 0)).$$

If $hf_K = f_* h$, the formula $[z,t] \mapsto [h(z),t]$ respects the seam and has the analogous inverse. For an isotopic target representative, the standard interpolating formula $[z,t] \mapsto [k_t(z),t]$, with $f_* k_1 = k_0 f_K$, gives the same conclusion. The alpha conjugacy is exact on the c collar. For beta, both sides use the identical supported twist word; interpolating twist profiles is relative to the disjoint e collar. The two handle maps restrict to h on their common fiber and therefore glue, as do their inverses. This constructs H directly; bundle classification and Dehn–Nielsen–Baer are independent controls, not proof inputs. □

Corollary 4.3 (the marked torus pair)**.** Under H, the induced subcomplexes $c \rtimes_{\varphi_K} S^1_\alpha$ and $e \times S^1_\beta$ map to T_α, T_β, together with their product collars and the p-section. All smooth and symplectic arguments may therefore be performed in the audit-defined smooth target R_{aud}; no smoothing of $|K|$ is chosen.

The model results through Corollary 4.3 concern V_{aud} and use no source-comparison clause. Five structurally different constructions test subdivision, monodromy order, twist sign, labels, and orientations. These internal controls support the audit object; they do not identify it with the object denoted by a source diagram. That open comparison is discussed only in Section 7.

5 Peripheral data and the product filling

Let $T = T_\alpha \sqcup T_\beta$. Transport the explicit normal block neighborhood $N_{1/2}(T)$ below through the marked homeomorphism H, and use the resulting product-collared neighborhood as $\nu(T)$. Put $C_{\mathrm{aud}} = R_{\mathrm{aud}} \setminus \operatorname{int} \nu(T)$. This section identifies the normal boundary of each component, its commonly based peripheral pair, and the two filling slopes. The statements are deliberately separated so that a failure of one cannot be hidden by triviality of the final groups. This section proves the intrinsic product-filling theorem; it uses no symplectic or Lagrangian-framing input.

5.1 Normal frontier and compact exterior

Lemma 5.1 (local flatness and the normal frontier)**.** The two torus subcomplexes are disjoint, full, and locally flat. The barycentric level set

$$N_{1/2}(T) = \left\{ x : \sum_{v \in T} \lambda_v(x) \geq \tfrac{1}{2} \right\}$$

is an explicit normal D^2-block bundle, and its frontier $F_{1/2}(T)$ is the normal-circle boundary. The derived frontier used by the computation is PL homeomorphic to $F_{1/2}(T)$, carrying every extracted dual loop to a literal oriented normal-circle fiber.

Proof. For every simplex $\sigma \subset T$, the complete link pair is checked to be

$$(S^3, \text{unknotted } S^1), \quad (S^2, S^0), \quad (S^1, \varnothing)$$

in dimensions $0, 1, 2$, respectively. The enumeration covers every simplex of both tori. Vertex-link 3-spheres and unknotted link circles carry independently replayed collapse and cyclic-knot witnesses. Since the star pair of a simplex is the join of that simplex with its link pair, these standard links give the required standard local chart at every point. This is the simplex-link formulation of PL local flatness; see [7, Chapter 4, pp. 50–51].

In a simplex with torus face A and opposite face B, barycentric coordinates give the unique block form

$$N_{1/2}(T) \cap \sigma \cong \Delta_A \times C(\Delta_B), \qquad F_{1/2}(T) \cap \sigma \cong \Delta_A \times \Delta_B.$$

The formulas agree on common faces. If $b(s)$ is a mixed-simplex vertex of the computed derived frontier, the map

$$b(s) \longmapsto \tfrac{1}{2} b(s \cap T) + \tfrac{1}{2} b(s \smallsetminus T)$$

extends cellwise to the canonical subdivision map for $\Delta_A \times \Delta_B$ and hence gives the global PL homeomorphism. Every mixed vertex and derived comparability is checked. For every torus triangle the alternating dual loop has constant torus coordinate and is precisely the subdivided normal link circle. No regular-neighborhood uniqueness or smoothing theorem is used. □

Lemma 5.2 (complement presentation)**.** Let $L = K[V(K) \smallsetminus V(T)]$ be the full subcomplex on the vertices outside T. There is a homotopy-equivalence diagram

$$|L| \hookrightarrow |K| \smallsetminus \operatorname{int} N_{1/2}(T) \hookrightarrow |K| \smallsetminus |T|, \qquad |K| \smallsetminus \operatorname{int} N_{1/2}(T) \cong C_{\mathrm{aud}},$$

in which both spaces on the right strongly deformation retract onto $|L|$. Consequently the maximal-tree edge generators of L and its oriented triangle boundaries give a presentation of $\pi_1(C_{\mathrm{aud}})$. A sealed Tietze certificate reduces the raw cellular presentation to the three-generator, seventy-eight-relator presentation Q of Theorem 3.1, while transporting all named peripheral and comparison words. Exact cellular and transport counts are in the supplement.

Proof. On each simplex not contained in the full subcomplex T, let $r(x)$ be the point obtained by setting the coordinates on T to zero and renormalizing the remaining barycentric coordinates. The straight-line homotopy $H_t(x) = (1-t)x + tr(x)$ agrees on faces and fixes $|L|$. If $\rho(x) = \sum_{v \in T} \lambda_v(x)$, then $\rho(H_t(x)) = (1-t)\rho(x)$. Thus H_t stays in $|K| \smallsetminus |T|$ and also restricts to the block exterior $\{\rho \leq 1/2\} = |K| \smallsetminus \operatorname{int} N_{1/2}(T)$. The homeomorphism H and the chosen product collar identify that block exterior with C_{aud}. This proves all three asserted homotopy identifications, including the previously implicit passage from the deleted subcomplex to the compact torus exterior.

Collapsing a maximal tree and applying cellular van Kampen to every oriented two-simplex gives the raw presentation. Each recorded Tietze elimination is replayed against the unique current relator that implies it; separate Python and Ruby checker implementations agree on the final presentation and every tracked word. □

Based meridians and product-framing longitudes for both tori are traced as literal simplicial loops carrying the audit definition's basing whiskers. External consistency checks compare the resulting words with Wuebben's corrected relation sheet and basing convention, but those comparisons play no logical role in any theorem about V_{aud}. Internally, a combinatorial sweep of the relevant transport square resolves the basing double-count question from the audit paths themselves. Both surgery tori are certified locally flat at every simplex link, and every extracted dual meridian is checked to be a literal normal-circle fiber of an explicit normal block model, with no regular-neighborhood theorem invoked. The doubled-section datum used in the historical target application is kept in the supplement; the intrinsic section of V_{aud} is treated directly below.

5.2 Relation sheet and product fillings

To make Theorem 5.9 inspectable without opening the repository, fix the orientation sheet used by the audit triangulation and put $\delta = r^{-1}$. Before filling, the resulting coordinate presentation read from the literal paths has the surface relation, the following eight transport relations, and the drilled-fiber relation:

$$\begin{aligned} AxA^{-1} &= r, & AyA^{-1} &= s, & ArA^{-1} &= x, & AsA^{-1} &= Ny, \\ BxB^{-1} &= y^{-1}, & ByB^{-1} &= M^{-1}yx, & BrB^{-1} &= r, & BsB^{-1} &= (r^{-1}M^{-1}r)s, \\ [x,y][r,s] &= 1, & B(s^{-1}r^{-1}yx)B^{-1} &= r^{-1}s^{-1}x. \end{aligned}$$

The two commonly based product-framed directions are

$$\lambda_\alpha = Ax, \qquad \lambda_\beta = (r^{-1}Mr)B,$$

and the four algebraic sign sheets append exactly

$$M\lambda_\alpha^{\epsilon_A} = 1, \qquad N\lambda_\beta^{\epsilon_B} = 1, \qquad (\epsilon_A, \epsilon_B) \in \{\pm 1\}^2. \tag{1}$$

Proposition 5.3 (abelianization of a slope family)**.** For $p, q \in \mathbb{Z}$, let $P_{p,q}$ be the quotient of the sealed complement presentation Q obtained by adjoining

$$M\lambda_\alpha^p = 1, \qquad N\lambda_\beta^q = 1.$$

With the convention $\mathbb{Z}/0 = \mathbb{Z}$,

$$H_1(P_{p,q}; \mathbb{Z}) \cong \mathbb{Z}/q \oplus \mathbb{Z}/p.$$

In particular the complement has infinitely many fillings with nontrivial fundamental group. Omitting either filling relation leaves an infinite cyclic summand, while $|p| > 1$ or $|q| > 1$ produces nonzero torsion in abelianization.

Proof. Let g_1, g_2, g_3 be the generators of Q in its sealed order. The exponent vectors of its 78 relators span exactly $\mathbb{Z}(0,0,1) \subset \mathbb{Z}^3$, so

$$H_1(Q; \mathbb{Z}) = \mathbb{Z}\langle g_1 \rangle \oplus \mathbb{Z}\langle g_2 \rangle.$$

The hash-bound tracked words have exponent vectors

	M	N	λ_α	λ_β
vector in $\mathbb{Z}^3$	$(0,0,1)$	$(0,0,0)$	$(0,1,2)$	$(-1,0,-3)$.

Thus their classes in $H_1(Q)$ are respectively $0, 0, g_2, -g_1$. The two new relations impose $pg_2 = 0$ and $qg_1 = 0$, which gives the displayed answer. The single-filling and nonunit-coefficient assertions follow immediately. □

These are four algebraic specializations in one fixed symbolic basis. Only the $(+,+)$ specialization is used below as a presentation of V_{aud}. The finite presentations of Theorem 3.1 are the frozen Tietze images of all four literal word lists; for the other three, no identification with a filled geometric manifold is asserted here.

Lemma 5.4 (extension across a torus filling)**.** Let $X = T^2 \times D^2$ and order an integral basis of its boundary as $(\mu, \gamma_1, \gamma_2)$, where μ bounds the disk fiber. A boundary diffeomorphism $f : \partial X \to \partial X'$ extends over X if and only if $f_*(\mu) = \epsilon\mu'$ for $\epsilon \in \{\pm 1\}$. In these bases its matrix then has the form

$$[f_*] = \begin{pmatrix} \epsilon & u_1 & u_2 \\ 0 & a_{11} & a_{12} \\ 0 & a_{21} & a_{22} \end{pmatrix} = \begin{pmatrix} \epsilon & u \\ 0 & A \end{pmatrix}, \qquad A \in GL(2,\mathbb{Z}), \quad u \in \mathbb{Z}^2.$$

Consequently a diffeomorphism of torus exteriors that carries each killed attaching slope to the corresponding slope up to sign extends across the filled pieces; the two surviving boundary directions are recorded by A and u.

Proof. Necessity follows because the kernel of $H_1(\partial X) \to H_1(X)$ is the primitive subgroup generated by μ. For sufficiency, write $x \in \mathbb{R}^2/\mathbb{Z}^2$ for the base coordinate and $z \in D^2 \subset \mathbb{C}$ for the disk coordinate. Up to translations, the displayed matrix is realized by

$$(x, z) \longmapsto \begin{cases} (Ax, e^{2\pi i \langle u,x\rangle} z), & \epsilon = +1, \\ (Ax, e^{2\pi i \langle u,x\rangle} \overline{z}), & \epsilon = -1. \end{cases}$$

This restricts to the required affine isotopy class on the boundary. The linearization theorem for the flat three-torus [8, case (IV), pp. 464–465] says that every diffeomorphism of T^3 is isotopic to an affine representative; its linear part is the induced action on integral first homology. Extending that isotopy through a boundary collar adjusts the displayed extension to f.

The orientation sign of the displayed four-dimensional map is $\epsilon \det A$, which is also $\det[f_*]$ on the ordered boundary basis. Thus an orientation-preserving boundary map has an orientation-preserving extension, and an orientation-reversing boundary map has an orientation-reversing extension. In the filled-exterior application one conjugates the exterior boundary map by the two attaching maps. If both attaching maps reverse the boundary orientation, their two signs cancel, so the conjugated map has the same orientation sign as the exterior map. This reduces the extension problem to the same meridian criterion and accounts explicitly for the third boundary direction as well as the orientation. □

Corollary 5.5 (peripheral convention changes)**.** Let (μ, λ, ν) be an oriented integral basis of the boundary three-torus of one deleted torus, with geometric filling slope $s = \mu + \lambda$. Reversing the orientations used to name both μ and λ, while leaving ν fixed, is the coordinate change

$$D_- = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in SL(3,\mathbb{Z}).$$

It carries s to $-s$ and hence preserves the unoriented filling slope. The corresponding boundary map extends across the filling $T^2 \times D^2$. In the commonly whiskered peripheral subgroup, the same relation is written

$$(\mu\lambda)^{-1} = \mu^{-1}\lambda^{-1} = 1.$$

By contrast, reversing only the displayed longitude changes the slope to $\mu - \lambda$; the matrix $\mathrm{diag}(1, -1, -1) \in SL(3,\mathbb{Z})$ sends s to that distinct primitive class. It is not a convention change preserving the fixed filling unless an additional self-diffeomorphism of the complement carrying $\mu + \lambda$ to $\mu - \lambda$ is supplied.

Proof. The two displayed matrices give the stated changes on $H_1(T^3;\mathbb{Z})$. Since s and $-s$ define the same primitive kernel, Lemma 5.4 extends the first coordinate change over the filling piece. Because the meridian and longitude use a common whisker, their boundary subgroup is abelian; hence inversion of $\mu\lambda$ gives the displayed word. Finally $(1,-1,0)$ is not $\pm(1,1,0)$, proving the last assertion. The same argument applies independently on the alpha and beta boundary components. $\square$

Direction	Based word	Geometric construction and evidence
λ_α	Ax	y_1-based product push-off at c_y; literal path construction and complement-only certificate
λ_β	$(r^{-1}Mr)B$	s_2-based product push-off at s_e; punctured transport square and complement-only certificate

Table 2: The two sign-sensitive longitude identifications. Neither certificate contains a filling relation. The words are oriented in the fixed audit basis; Lemma 5.5 states the complete coordinate change required to preserve the filling.

5.3 Peripheral curves and drilled transport

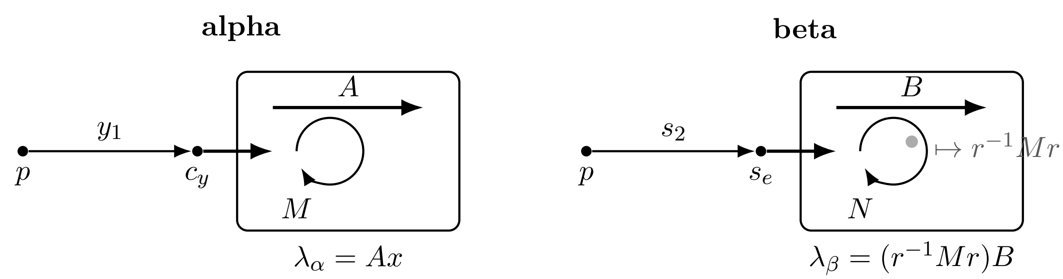


Figure 3: Common-whisker peripheral construction. On the alpha side the meridian and longitude use the same y_1 path to c_y. On the beta side they use the same s_2 path to s_e; the swept basing square meets T_α once, and puncturing that intersection contributes the conjugated meridian $r^{-1}Mr$. The diagram records basing and word order, not metric geometry.

Lemma 5.6 (the alpha peripheral pair)**.** Under the normal-frontier identification of Lemma 5.1, the extracted loops are the commonly based meridian and product longitude (M, λ_α), and

$$\lambda_\alpha = Ax \quad \text{in } \pi_1(C_{\mathrm{aud}}).$$

The displayed equality is certified using only the 78 complement relators, with neither filling relation present.

Proof. Consider T_α. The curve c meets the oriented y edge once, at c_y. Starting at p, the stored path follows the initial segment y_1 to c_y, makes a normal jog to the frontier of the explicit normal block, and traverses the oriented dual normal circle. Returning along the same whisker gives M. The longitude uses that identical whisker and normal jog, traverses the base direction A, and closes along that same half of c, from $c_s = \varphi_0(c_y)$ to c_y. These two literal based paths are what M and λ_α denote throughout and what the finite presentation fills with.

That the two loops use the same whisker is not an inference: in the sealed transport input they are $y_1\mu_\alpha y_1^{-1}$ and $y_1\lambda_\alpha^\circ y_1^{-1}$ with the identical whisker y_1. So no independent conjugation is hidden in the filling word $M\lambda_\alpha^{\epsilon_A}$.

The identification of the extracted longitude with Ax has two independent supports. Geometrically it follows from how the path is built — based on the y rail at the unique crossing c_y, whiskered by y_1, jogged to the normal-block frontier, lifting the base generator A, and closed along the half of c from c_s to c_y. Algebraically, the elementary elimination pass over the sealed presentation leaves a nonempty residual for $\lambda_\alpha^{-1}Ax$, but a proof-producing completion reduces that exact word to the identity. Independent Python and Ruby verifiers replay its complete ancestry cone from the sealed three-generator, 78-relator complement; neither filling relation is present. The supplement gives the exact path identifiers, word lengths, certificate inventory, and frozen source locations. □

Lemma 5.7 (the beta peripheral pair)**.** Under the same normal-frontier identification, the extracted loops are the commonly based meridian and product longitude (N, λ_β), and

$$\lambda_\beta = (r^{-1}Mr)B \quad \text{in } \pi_1(C_{\mathrm{aud}}).$$

Again the equality is certified in the unfilled complement.

Proof. The beta-coordinate word $\lambda_\beta^{-1}(r^{-1}Mr)B$ likewise leaves a nonempty residual under elementary elimination. The same proof-producing complement system reduces that exact word to the identity; separate Python and Ruby verifier implementations replay its complete ancestry cone. Thus both displayed coordinate identities are algebraically certified in the complement, although Theorem 3.1 remains more direct: it fills with the literal stored boundary longitudes rather than relying on either coordinate substitution.

The beta extraction is the sign-sensitive case. The corresponding path uses the initial segment s_2 to the unique crossing s_e, the base loop B, and the fiber-normal push-off. Its swept basing square crosses T_α once. Puncturing that square contributes the conjugated meridian $r^{-1}Mr$, which is why the audit word for this direction is $(r^{-1}Mr)B$ and not B. The extracted paths are exported without coordinate substitution; they are what N and λ_β denote, and the identification with $(r^{-1}Mr)B$ is certified in the complement exactly as in the α case. An independently written extractor recovers both pairs without importing the complement, sweep, or peripheral modules. Local link checks certify both tori at every simplex, and the explicit block-frontier map checks every dual loop as a normal-circle fiber; these are the finite facts behind the geometric words “meridian” and “push-off.” □

Lemma 5.8 (the drilled-fiber relation)**.** The based relation

$$B(s^{-1}r^{-1}yx)B^{-1} = r^{-1}s^{-1}x$$

holds in $\pi_1(C_{\mathrm{aud}})$, not merely in the unpunctured mapping cylinder.

Proof. The transported loop κ_3 misses c but meets e once; sweeping it along the core of the beta curve would therefore meet T_β and would not prove the stated complement relation. The audit uses the specified parallel push-off. Along that push-off, the complete mapping-cylinder stack has no vertex on either surgery torus. A simplicial homotopy grid at p joins the push-off basepoint loop to the core one and identifies it with the presentation's whiskered generator; it too is disjoint from both tori. The based monodromy target then reduces to $r^{-1}s^{-1}x$.

The finite checker exhausts the stack, both torus vertex sets, every grid cell, and the target path. Thus the entire transport homotopy lies in C_{aud}. This is the construction-specific fact that was absent from the earlier mapping-cylinder-only calculation; no group simplification or low-index fingerprint is used to supply it. $\square$

5.4 Simple connectivity of the product-filled audit model

Theorem 5.9 (audit-model identification)**.** For the fixed audit sheet $(\epsilon_A, \epsilon_B) = (+1, +1)$ the natural map is an isomorphism

$$P_{+,+} \;\cong\; \pi_1(V_{\mathrm{aud}}).$$

No geometric identification of $P_{+,-}, P_{-,+}, P_{-,-}$ is part of this statement. No clause of Source Comparison Hypotheses D1–D14 enters this theorem.

Proof. Lemmas 4.1 and 4.2 construct the marked bundle homeomorphism and carry the two curve subbundles and their product collars to the specified tori. Lemma 5.1 identifies their normal boundaries and meridians. Lemmas 5.6 and 5.7 identify the two commonly based product longitudes, including the conjugated meridian contributed by the punctured beta sweep. Lemma 5.8 places the final transport relation in the drilled complement.

By Definition 2.1, attaching the two product-framed filling pieces adds exactly the normal closure

$$\langle\!\langle M\lambda_\alpha, N\lambda_\beta \rangle\!\rangle.$$

Cellular van Kampen therefore gives a surjection from the filled presentation. Lemma 5.2 gives an actual presentation of the complement, not merely a list of valid relations; applying van Kampen with the two filling relations upgrades that surjection to the asserted isomorphism. Changing a common whisker simultaneously conjugates both elements of a peripheral pair and does not change their normal closure. Lemma 5.5 records the different, simultaneous meridian–longitude reversal that preserves a filling slope. This completes the identification with Definition 2.1 without using a symplectic form, a Weinstein chart, or any comparison with Wuebben's text and figures. $\square$

Proof of Theorem 2.2, except symplecticity. Use the $(+,+)$ sheet in Theorem 5.9. Its source is trivial by Theorem 3.1; a quotient of the trivial group is trivial. Notice that this conclusion needs only surjectivity in Theorem 5.9, no symplectic or Lagrangian-framing result, and no source-identification assumption.

The boundary of R_{aud} is the F-bundle over $\partial\Sigma_{1,1} \cong S^1$ whose monodromy is the commutator determined by the oriented boundary word in the two clutching maps. It is connected. Both torus surgeries take place in the interior, so this is also $\partial V_{\mathrm{aud}}$. Since

$$\chi(F) = 2 - 2\cdot 2 = -2, \qquad \chi(\Sigma_{1,1}) = 2 - 2 - 1 = -1,$$

multiplicativity for the bundle gives $\chi(R_{\mathrm{aud}}) = 2$. Removing and regluing two copies of $T^2 \times D^2$, each with Euler characteristic zero and boundary T^3, does not change Euler characteristic. Hence $\chi(V_{\mathrm{aud}}) = 2$.

Simple connectivity gives $H_1(V_{\mathrm{aud}}; \mathbb{Z}) = 0$. Because both the manifold and its boundary are connected, the beginning of the cohomology sequence of the pair, together with $H^1(V_{\mathrm{aud}}; \mathbb{Z}) =$

0, gives $H^1(V_{\mathrm{aud}}, \partial V_{\mathrm{aud}}; \mathbb{Z}) = 0$. Lefschetz duality therefore gives $H_3(V_{\mathrm{aud}}; \mathbb{Z}) = 0$. Similarly the homology sequence of the pair gives $H_1(V_{\mathrm{aud}}, \partial V_{\mathrm{aud}}; \mathbb{Z}) = 0$; duality and the universal coefficient theorem then show that $H_2(V_{\mathrm{aud}}; \mathbb{Z})$ is torsion-free. These are the cases of the universal coefficient theorem and Lefschetz duality stated in [9, Theorems 3.2 and 3.43]. Since the boundary is nonempty, $H_4 = 0$, and the Euler characteristic now forces $H_2 \cong \mathbb{Z}$.

Choose a base point in the interior of the base away from the two clutching loops. Its fiber F misses both surgery tori, survives the fillings, and has trivial normal bundle, hence $F^2 = 0$. The fixed point p, which lies on neither c nor e, gives a proper section $\Gamma \cong \Sigma_{1,1}$ disjoint from the surgery tori. It also survives, and $F \cdot \Gamma = 1$. Thus $[F]$ is primitive and generates H_2. Modulo two, H_2 is generated by $[F]$. On F the oriented splitting $TV_{\mathrm{aud}}|_F = TF \oplus \nu F$, the Whitney product formula, and the mod-two reduction of the two Euler classes give

$$\langle w_2(V_{\mathrm{aud}}), [F] \rangle = \chi(F) + F^2 = -2 + 0 = 0 \pmod 2.$$

See [10, §§4 and 9]. Since evaluation on $[F]$ detects $H^2(V_{\mathrm{aud}}; \mathbb{Z}/2)$, it follows that $w_2(V_{\mathrm{aud}}) = 0$, so V_{aud} is spin. The remaining symplectic assertion is proved after Theorem 6.3. □

6 The product-to-Lagrangian framing bridge

The proof of simple connectivity and the intrinsic topological assertions is complete before this section begins. The framing theorem is not used in Theorem 5.9. It proves the source-independent symplectic assertion of Theorem 2.2; it is also the relevant bridge in any future comparison with a Lagrangian source construction.

Lemma 6.1 (bundle symplectic form and protected collars)**.** The monodromy representatives in Definition 2.1 may be chosen to preserve one positive area form Ω on F, with ψ_0 equal to the identity on a collar of e and φ_0 equal to the stated half-rotation on a collar of c. Write $\pi : R_{\mathrm{aud}} \to \Sigma_{1,1}$ for the bundle projection. If η is a positive area form on $\Sigma_{1,1}$, then for every $K > 0$

$$\omega_K = \Omega_{\mathrm{vert}} + K\pi^*\eta$$

is a closed symplectic form on R_{aud}. The two tori T_α, T_β are Lagrangian. On disjoint protected neighborhoods of them there are oriented coordinates $(\theta_1, t, \theta_2, u)$ in which

$$\omega_K = f(\theta_1, t)\, dt \wedge d\theta_1 + K\, du \wedge d\theta_2, \qquad f > 0,$$

with no mixed vertical–horizontal terms. On the alpha neighborhood these coordinates obey the seam $(\theta_1, t, 2\pi, u) \sim (\theta_1 + \pi, t, 0, u)$; on the beta neighborhood they are ordinary product coordinates.

Proof. Average any positive area form over the order-two action of φ_0. An equivariant tubular coordinate on the invariant curve c then writes $\varphi_0(\theta, t) = (\theta + \pi, t)$ and the averaged form as $f(\theta, t) dt \wedge d\theta$. For this same area form, represent T_a and T_b by the usual area-preserving annular shears. Because e is disjoint from a and b, their supports may be chosen disjoint from a protected collar of e; hence $\psi_0 = T_a \circ T_b$ preserves Ω and is the identity on that collar. Thus both clutching maps preserve one area form and retain the required collar behavior.

Use the flat product atlas determined by the two locally constant clutching maps. The fiber form descends in that atlas to a globally defined closed vertical form Ω_{vert}: on overlaps the transition is locally constant in the base and preserves Ω. Therefore $d\omega_K = 0$, and, with the fiber and base orientations fixed by Ω and η,

$$\omega_K^2 = 2K\,\Omega_{\text{vert}} \wedge \pi^*\eta > 0.$$

This proves symplecticity and fixes the total orientation. Each surgery torus has one fiber tangent and one base tangent, so both summands restrict to zero on it.

Choose $\eta = du \wedge d\theta_2$ on disjoint annular collars of the two base loops and write $\Omega = f(\theta_1, t)dt \wedge d\theta_1$ on the protected fiber collars. The flat atlas then gives the displayed split expressions; there are no mixed terms to remove. The identity behavior of ψ_0 gives the beta product chart, while the half-rotation of φ_0 gives exactly the alpha seam. These choices can be made simultaneously because the protected collars are disjoint. We orient the normal momentum plane by (t, Ku); its boundary orientation is the positive meridian convention used below. $\square$

Definition 6.2 (Lagrangian-framing class)**.** Let T be either audit torus with its oriented product coordinates, and let $\gamma \subset T$ be one of the named primitive curves. In a Weinstein chart fixing the zero section, take a sufficiently small nonzero constant covector copy of γ and radially project it to $\partial\nu(T)$. Its isotopy class in $\nu(T) \smallsetminus T$ is the chartwise Lagrangian push-off class. The chart-independence part of Theorem 6.3 proves that this class is independent of the Weinstein chart and of the sufficiently small constant covector; the resulting boundary class is what we call the canonical Lagrangian-framing class of γ.

Theorem 6.3 (product framing equals Lagrangian framing)**.** For the symplectic form ω_K of Lemma 6.1, the product-framed push-offs in Lemmas 5.6 and 5.7 represent the canonical Lagrangian-framing classes of T_α and T_β. In particular neither acquires a meridian component when transported into a Weinstein neighborhood.

Proof. The proof takes place in the already smooth target R_{aud}. By Corollary 4.3, the marked homeomorphism H transports the product collars and their push-offs from the PL model into that target. No smooth structure is placed on the source complex. A framing push-off below means the isotopy class, in $\nu(T_i) \smallsetminus T_i$, of a sufficiently small copy of a curve on T_i, as in Definition 6.2.

Relative normal form. On an annular fiber collar write

$$\Omega = f(\theta, t)\,dt \wedge d\theta, \qquad f > 0,$$

with core $t = 0$. Put

$$g(\theta, t) = \int_0^t (f(\theta, u) - 1)\,du, \qquad H_s(\theta, t) = t + sg(\theta, t).$$

Then $\partial_t H_s = (1 - s) + sf > 0$. On one collar common to the compact core, H_s therefore has a smooth inverse in the radial coordinate. If $H_s(\theta, T_s(\theta, t_0)) = t_0$, then $T_0 = t_0$, $T_s(\theta, 0) = 0$, and

$$\partial_s T_s = -\frac{g(\theta, T_s)}{(1 - s) + sf(\theta, T_s)}.$$

Thus $(\theta, t_0) \mapsto (\theta, T_s)$ is the relative Moser isotopy, obtained by monotone inversion rather than an abstract flow theorem. At $s = 1$, differentiation in t_0 gives $f(\theta, T_1)\partial_{t_0} T_1 = 1$, hence

$T_1^*\Omega = dt_0 \wedge d\theta$. The exact identities, fixed-core condition, and a uniform collar bound have a finite symbolic replay listed in the supplement. Apply this construction independently on the disjoint protected c and e collars. It therefore standardizes both neighborhoods simultaneously while fixing both torus cores and leaving the form unchanged outside slightly larger disjoint collars.

For the alpha collar the deck involution is $\tau(\theta, t) = (\theta + \pi, t)$ and the quotient coordinate is $\bar{\theta} = 2\theta$; write π_α for the quotient map. The preceding radial map is defined directly upstairs by replacing $\bar{\theta}$ with 2θ in its coefficient. Periodicity makes it τ-equivariant, so no covering-lift choice is involved. Moreover

$$\pi_\alpha^*(dt \wedge d\bar{\theta}) = 2\, dt \wedge d\theta = d(2t) \wedge d\theta,$$

which is the required factor-two normalization. The corresponding equivariance identities are replayed in the supplement's framing inventory.

The two charts. Near T_β, ψ_0 is the identity near e, and the preceding normalization gives

$$\omega = dt \wedge d\theta_1 + K\, du \wedge d\theta_2, \qquad K > 0;$$

the cotangent momenta are (t, Ku). The in-fiber and base-normal push-offs are therefore constant-momentum copies. Near T_α, the mapping-torus seam is

$$(\theta_1, t, 2\pi, u) \sim (\theta_1 + \pi, t, 0, u).$$

On its quotient put

$$\Theta_1 = \theta_1 - \tfrac{1}{2}\theta_2, \quad \Theta_2 = \theta_2, \quad P_1 = t, \quad P_2 = \tfrac{1}{2}t + Ku.$$

These functions respect the seam and satisfy

$$dP_1 \wedge d\Theta_1 + dP_2 \wedge d\Theta_2 = dt \wedge d\theta_1 + K\, du \wedge d\theta_2.$$

The fiber push-off has momentum $(t_0, t_0/2)$ and the half-drifting base push-off has momentum $(0, Ku_0)$, both constant. Hence neither acquires a meridian component, which is precisely the equality of framings used in (1). The Liouville primitive $t\, d\theta_1 + Ku\, d\theta_2$ forces these momenta by pairing with the Θ-frame; the supplement records an independent replay of that derivation, the seam, exterior algebra, and a second opposite-shear chart.

$$\boxed{\begin{array}{c}T_\beta \text{ coordinates}\\ (\theta_1,\theta_2;t,Ku)\end{array}} \xrightarrow{\text{product}} \boxed{\begin{array}{c}\text{constant momenta}\\ \text{fiber: } (t_0,0) \quad \text{base-normal: } (0,Ku_0)\end{array}}$$

$$\boxed{\begin{array}{c}T_\alpha \text{ seam coordinates}\\ (\theta_1,\theta_2;t,u)\end{array}} \xrightarrow{\text{shear}} \boxed{\begin{array}{c}\text{cotangent coordinates}\\ \Theta_1=\theta_1-\tfrac{1}{2}\theta_2,\ P_1=t\\ \Theta_2=\theta_2,\ P_2=\tfrac{1}{2}t+Ku\end{array}}$$

$$\begin{array}{c}\text{fiber push-off: } (P_1,P_2)=(t_0,t_0/2)\\ \text{half-drift push-off: } (P_1,P_2)=(0,Ku_0)\end{array}$$

Figure 4: The two local Weinstein-coordinate calculations. Product push-offs become constant-covector copies in both charts, so the path from a push-off to the torus never winds in a normal circle. The chart-independence argument below shows that this boundary framing class is intrinsic.

Here is the boundary-class assertion in coordinates. Let γ be either named curve, let η complete it to an oriented basis of the zero-section torus, and let μ be the positively oriented circle in a cotangent fiber. After radial projection to the unit-covector boundary, a constant nonzero covector p_0 gives

$$s \longmapsto \big(\gamma(s), p_0/\|p_0\|\big).$$

Its normal-circle coordinate is constant, so its class in the boundary basis (γ,η,μ) has μ-coefficient zero. This is the precise sense in which "constant covector" excludes a meridional component. The seam map and the quotient map above preserve the displayed orientations: their pullbacks give the positive forms written there, and the chosen positive normal circle is the boundary orientation of the ordered momentum plane.

Independence of the Weinstein chart. It remains to show that the constant-momentum conclusion is a framing class rather than a feature of the displayed coordinates. Given two Lagrangian-neighborhood charts, first compose with a cotangent lift so that their restrictions to the zero section agree. Their transition is then a symplectomorphism germ $\Phi : (T^*T, T) \to (T^*T, T)$ fixing the zero section pointwise. With fiber dilation $\delta_r(q,p) = (q, rp)$, define

$$\Phi_r = \delta_r^{-1}\Phi\delta_r \quad (0 < r \le 1).$$

Although $\delta_r^*\omega = r\omega$, the two factors cancel, so every Φ_r is symplectic. Along the zero section the symplectic condition gives

$$d\Phi = \begin{pmatrix} I & A \\ 0 & I \end{pmatrix}, \qquad A = A^{\mathrm{T}}.$$

Taylor expansion shows that Φ_r extends at $r = 0$ by the identity. Hence Φ_r is an isotopy of germs, relative to T, from the identity to the chart transition. To make the punctured-neighborhood assertion uniform, choose a compact germ domain over the entire named curve and then choose one nonzero constant covector copy small enough that every Φ_r is defined on it. Each Φ_r fixes the zero section and is injective, so no point of that copy can meet the zero section. Compactness of the copy times $[0,1]$ then gives a positive distance from the zero section throughout the isotopy. Thus the whole isotopy lies in one $\nu(T) \smallsetminus T$ and proves that the two push-offs have the same boundary class. The first-jet symplectic calculation is replayed by the checker listed in the supplement. The potentially dangerous momentum shift by a closed one-form could alter the pushed-off boundary class, but it moves the zero section unless the form vanishes and is therefore excluded here. The Lagrangian-neighborhood statement in the first paragraph of [5, §2.1], applied to each compact torus, supplies the required zero-section Weinstein charts; the argument above supplies the precise chart-independence consequence needed here. □

Completion of the proof of Theorem 2.2. Lemma 6.1 makes the two disjoint audit tori Lagrangian, and Theorem 6.3 identifies the two product longitudes in Definition 2.1 with their Lagrangian-framing classes. The two unit-coefficient fillings are therefore Luttinger surgeries in the sense of [5, Definition 2.1]; the symplectic form on the filled manifold is the one asserted in [5, Proposition 2.2]. That construction supplies a symplectic form after each surgery which agrees with ω_K outside the chosen torus neighborhood. Performing the surgeries successively gives a symplectic form on V_{aud}. The fiber F chosen in the proof above is disjoint from both surgery neighborhoods, so the resulting form still restricts positively to F. □

7 Relation to Wuebben's construction and review boundary

Nothing in Theorem 2.2 identifies V_{aud} with Wuebben's fixed surgery member. The companion supplement records fourteen comparison clauses D1–D14: textual and diagrammatic clauses D1–D11 specify the proposed dictionary, while D12–D14 ask for the actual relative smooth identification of the two Lagrangian tori, framing conventions, and marked bundles. Those clauses are a checklist for future comparison, not hypotheses of any theorem in this article. In particular, no exotic-manifold conclusion is asserted here.

7.1 The reported contrary computation and live failure locations

Wuebben's public repository reports that Lidman and Piccirillo obtained a fundamental-group calculation disagreeing with his, currently summarized as $\pi_1(V) \neq 1$. Their published paper [4] proves $H_1(V) = 0$ and normal generation by $\pi_1(F)$ but does not publish this contrary presentation. No relation sheet, generator dictionary, or nontriviality witness for the reported calculation is available in the materials examined here.

The finite Theorem 3.1 is unaffected because its objects are explicit strings. If the contrary calculation concerns the same geometric member, the live explanations are deliberately left neutral: a source-to-audit clause may fail; a displayed geometric identification or finite checker may be defective; the contrary calculation may be defective; or the computations may concern different members or conventions. A relation-by-relation reconciliation requires the missing presentation, dictionary, and witness. Until that material exists, this article makes no claim about Wuebben's V.

7.2 Independent review targets

Three kinds of independent review remain valuable. First, both theorem-critical filled-group checkers arose within the same AI-assisted development process. Separate languages, negative controls, and the mathematical specification reduce ordinary software risk, but do not replace an external conformance audit of one checker.

Second, the principal geometric risk for simple connectivity lies in Sections 4–5, not in the certificate arithmetic. A PL reviewer should check the equivariant marked-fiber extension, relative clutching, simplex-link local-flatness conversion, normal-frontier equivalence, both commonly based peripheral words, and the claim that the drilled transport homotopy stays in the complement. Those are the steps that turn the finite presentation into $\pi_1(V_{\mathrm{aud}})$.

Third, a symplectic reviewer should check that the split form in Lemma 6.1 descends and standardizes both collars, that the seam and normal-circle orientations agree with the filling basis, that constant-covector copies have zero meridional coefficient, and that the chart-independence isotopy remains in a uniform punctured neighborhood. The finite checks localize these questions; they do not replace specialist review. The supplement gives exact files and a longer review checklist.

The 100-case framing scan is not evidence for the dictionary: every sampled shifted presentation was reported trivial. By contrast, Proposition 5.3 supplies a transparent negative control showing that infinitely many other slope fillings have nontrivial abelianization.

8 Conclusion

Four manifest-bound presentations are trivial under a complete certificate specification. A separate geometric reconstruction identifies one of them with the explicitly defined V_{aud}, yielding the intrinsic topological statements of Theorem 2.2. The framing theorem then supplies its symplectic structure. This separation between finite derivation, geometric conversion, and symplectic framing is the article's source-independent contribution.

Whether Wuebben's text and diagrams specify the same marked smooth surgery data remains open. The supplement records that comparison and its historical downstream consequences without promoting either into a theorem of this article.

9 Data and reproducibility

The theorem artifacts, frozen inputs, certificate checkers, manifests, and reconstruction programs are in the public repository and its archival deposit [3]:

`https://github.com/VentiMath/exotic-s2xs2-verification/releases/tag/v2.2.2`,

`https://doi.org/10.5281/zenodo.22182187`.

The version DOI identifies the exact artifact base described here. The concept DOI `https://doi.org/10.5281/zenodo.22169753` resolves to the newest archived version.

The supplement gives the sealed-input digest, git tree object, literal replay commands, expected outputs, negative controls, and source-extraction pins. Certificate-only replay requires standard Python or Ruby; full reconstruction uses the recorded GAP 4.11.1 and KBMAG 1.5.9 environment. The theorem about V_{aud} replays from frozen files without locally supplied copies of either source paper.

References

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[3] J. Clyde, *A certificate-based audit of simple connectivity for an explicit model associated with Wuebben's proposed exotic $S^2 \times S^2$ construction*, verification artifact v2.2.2, Zenodo (2026), `https://doi.org/10.5281/zenodo.22182187`.

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